

● HARD

● ADVANCED MATH

Advanced Math

10 questions · ~15 min

15

SAT QUESTION BANK

2026-05-29

Q1

GRID-IN ADVANCED MATH

$$-x^2 + bx - 676 = 0$$

In the given equation, b is a positive integer. The equation has no real solution. What is the greatest possible value of b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A quadratic function models the height, in feet, of an object above the ground in terms of the time, in seconds, after the object is launched off an elevated surface. The model indicates the object has an initial height of 10 feet above the ground and reaches its maximum height of 1,034 feet above the ground 8 seconds after being launched. Based on the model, what is the height, in feet, of the object above the ground 10 seconds after being launched?

- A. 234
- B. 778
- C. 970
- D. 1,014

The function f is defined by $f(x) = (x+3)(x+1)$. The graph of f in the xy -plane is a parabola. Which of the following intervals contains the x -coordinate of the vertex of the graph of f ?

- A. $-4 < x < -3$
- B. $-3 < x < 1$
- C. $1 < x < 3$
- D. $3 < x < 4$

Q4

GRID-IN ADVANCED MATH

The product of two positive integers is 462. If the first integer is 5 greater than twice the second integer, what is the smaller of the two integers?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5

GRID-IN

ADVANCED MATH

The function g is defined by $gx = xx - 2x + 6^2$. The value of $g(7 - w)$ is 0, where w is a constant. What is the sum of all possible values of w ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The functions f and g are defined by the given equations.

$$fx = 3 + | -2x - x^2 |$$

$$gw = | \frac{-w}{w-1} | - w + 5$$

If $f(-4) = c$, where c is a constant, what is the value of gc ?

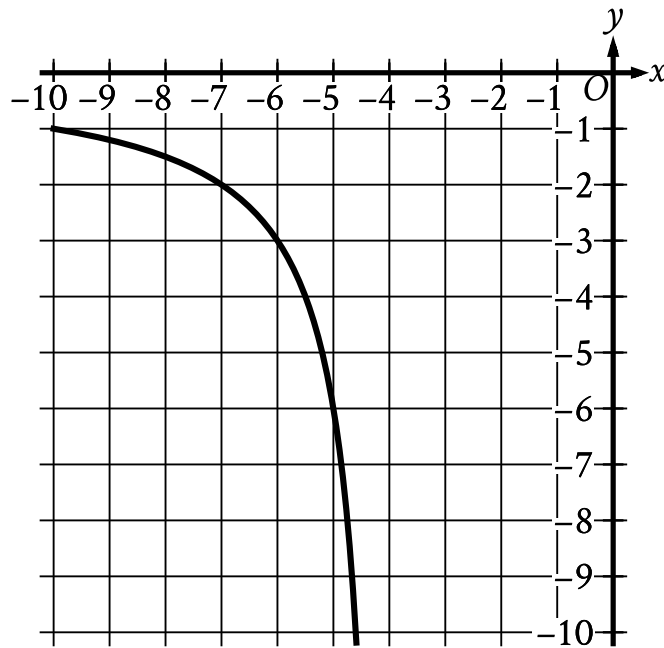
STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The function $f(x) = \frac{1}{9}x^2 - 7x + 3$ gives a metal ball's height above the ground $f(x)$, in inches, x seconds after it started moving on a track, where $0 \leq x \leq 10$. Which of the following is the best interpretation of the vertex of the graph of $y = f(x)$ in the xy -plane?

- A. The metal ball's minimum height was 3 inches above the ground.
- B. The metal ball's minimum height was 7 inches above the ground.
- C. The metal ball's height was 3 inches above the ground when it started moving.
- D. The metal ball's height was 7 inches above the ground when it started moving.



- Moving from left to right:
 - The curve is in quadrant 3.
 - The curve trends down sharply.
- As x increases, the curve approaches the line x equals negative 4.
- As x decreases, the curve approaches the line y equals 0.
- The curve passes through the following points:
 - (negative 7 comma negative 2)
 - (negative 6 comma negative 3)
 - (negative 5 comma negative 6)

The rational function f is defined by an equation in the form $fx = \frac{a}{x+b}$, where a and b are constants. The partial graph of $y = fx$ is shown. If $gx = fx + 4$, which equation could define function g ?

A. $gx = \frac{6}{x}$

B. $gx = \frac{6}{x+4}$

C. $gx = \frac{6}{x+8}$

D. $gx = \frac{6(x+4)}{x+4}$

$$fx = 4x^2 - 50x + 126$$

The given equation defines the function f . For what value of x does fx reach its minimum?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

The population of a town is currently 50,000, and the population is estimated to increase each year by 3% from the previous year. Which of the following equations can be used to estimate the number of years, t , it will take for the population of the town to reach 60,000 ?

A. $50,000 = 60,000(0.03)^t$

B. $50,000 = 60,000(3)^t$

C. $60,000 = 50,000(0.03)^t$

D. $60,000 = 50,000(1.03)^t$